Healthcare Digital Service Modernization Initiative: Agile Delivery Approach

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Summary

We want to update how a local healthcare provider handles its digital services using an Agile approach. Specifically, we are improving the way requests for CT scans, MRIs, and ultrasounds are processed. Our goal is to make sure these requests are tracked digitally and completed before strict deadlines. We are only focusing on the intake steps: submitting requests, checking them automatically, having doctors review them, updating their status, and keeping audit logs. We aren't touching things like billing, booking appointments, or writing the actual medical reports.We know the rules: privacy is a top priority, we must pass strict reviews before launching anything, and the medical rules will likely change as we work. Since doctors and staff are extremely busy, we will check in with them at scheduled times rather than demanding constant access. Because of all this, our plan is flexible and takes things step-by-step to ensure patient safety, get good feedback, and follow all the rules.

Part 1: Agile Delivery Approach Rationale

1.1 Proposed Agile Delivery Approach

For the intake process, our team is using a custom Agile method that combines Scrum, Kanban, and Extreme Programming (XP).

- Scrum gives us a schedule with short work cycles (sprints) and clear roles.
- Kanban uses visual boards to watch the work flow and stop jams before they happen.
- XP brings the strict coding standards needed to safely build medical software.
- Together, this layered approach lets us build things safely even when medical rules change.

1.2 Framework Comparison: Scrum, Kanban, and XP

It helps to look at why we combined these three methods.

1. Scrum is great for keeping things predictable, but it is hard to change direction in the middle of a sprint.
2. Kanban is awesome for visualizing a steady flow of work, but it lacks mandatory meetings to keep everyone on the same page.
3. XP is perfect for writing flawless code (which is a must in healthcare), but it doesn't offer much for high-level project planning.
4. By combining them, we get Scrum's structure, Kanban's visual tracking, and XP's quality—canceling out the weak spots of using just one method.

1.3 Work Organization and Management

We keep a main to-do list called a Product Backlog. The Product Owner writes tasks from the user's perspective and adds strict rules for what counts as "done". During our Sprint Planning meetings, developers grab the most important tasks to work on. We track these on a visual Kanban board under columns like "Pending," "Active," and "Review". Our Scrum Master runs daily check-ins to fix any problems, and we show our finished work to hospital staff during scheduled Sprint Reviews.

1.4 Justification for Key Delivery Decisions

We work in cycles because we can't possibly know every single requirement right at the start. Since hospital rules will change, flexible planning saves us from wasting time on rigid plans. We build the software in small pieces because hospital staff are too busy for constant updates. Showing them working software during our scheduled reviews gets us the best and most useful feedback.

1.5 Healthcare and Operational Context Fit

Our custom Agile setup fits perfectly with the healthcare industry's need to avoid risks. We build audit logs directly into our tasks, so following the rules happens naturally as we code. Scrum helps us manage complex steps, and its set review times fit perfectly with busy hospital schedules where constant, on-demand feedback just isn't possible.

1.6 Agile Practice Not Adopted

One common Agile practice we are skipping is "continuous deployment"—which is when code goes live automatically once it passes testing. In healthcare, strict rules say a formal privacy and security check must happen before anything goes live. So, we hold back our finished work in a controlled way until it passes these outside audits.

Part 2: Key Delivery Moments

2.1 Moment 1: Intake Workflow Discovery and Planning

In the beginning, our main job is to figure out the exact workflow steps and legal boundaries. The Product Owner breaks these down into a list of tasks. For us, being "ready to begin" simply means we have enough clear, important tasks—with privacy rules attached—to fill up our first work sprint.

2.2 Moment 2: Adjustment After Clinical Intake Rule Change

If a medical rule suddenly changes (for example, if urgent MRIs now need a specialist review), our setup keeps us from getting stuck. The Product Owner quickly writes up the new rule and puts it at the top of the list for the next sprint. For any older files, we use our audit logs to find them and have doctors review them again under the new rules. We talk about the change in our daily meeting so the developers working on the MRI features can adjust without stopping the whole team.

2.3 Moment 3: Readiness for Operational Review and Release

When we get close to launch, our focus shifts entirely to following the rules. We gather our audit logs and test patient records to prove we followed privacy laws during the review. The team double-checks that every single task meets our strict "Definition of Done". Being "ready for release" means the software works perfectly, the tracking logs are solid, and we passed the mandatory privacy checks.

Part 3: Evidence of Agile Values and Principles

3.1 Aligning Agile Manifesto Values to Team Actions

Here is how we use the core Agile values in our daily work:

- **People over processes:** We use a Kanban board, but we actually talk to each other directly to solve problems. When the MRI rule changed, we talked about it in our daily meeting instead of just quietly updating a ticket.
- **Working software over heavy documentation:** Medical software requires a ton of paperwork, so we programmed the system to create patient and audit records automatically as it runs, instead of making someone do it by hand.

- **Customer collaboration over contracts:** Since hospital staff are so busy, the Product Owner fights for what the clinic needs every single day. When we do meet with stakeholders, we look at working software instead of arguing over dry, boring documents.
- **Responding to change over following a plan:** We know rules will change. When they do, we easily adjust our upcoming to-do list instead of sticking to a broken, old plan.

3.2 Demonstration of Agile Principles

Our team also follows key Agile principles. We release working software in short bursts, letting busy administrators see real progress at our scheduled meetings. We also welcome late changes because our to-do list is flexible enough to handle them without messing up the whole project.

3.3 Upholding Agile Values Despite Constraints

Working in a strict medical field doesn't stop us from being Agile; we just have to be smart about it. We bake privacy rules right into our "Definition of Done" so compliance is never an afterthought. By pausing for audits instead of launching automatically, we stay flexible on the inside while still following the strict outside rule.

Conclusion

Upgrading a digital medical intake system is a balancing act between moving fast and following strict rules. By combining Scrum, Kanban, and XP, our team built a plan that can easily handle changing medical rules. We work in short sprints, make logging automatic, and respect doctors' time by only showing them finished, working software. Ultimately, this tailored strategy delivers a safe, rule-following system on time while keeping patient data completely secure.

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